ELEC70079
EE9-AO9
EE9-SO9

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

OPTICAL COMMUNICATION

Wednesday, 14 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer Question ONE, and any THREE of Questions 2 to 5.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : E.M. Yeatman
Second Marker(s) : S. Lucyszyn

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Special instructions for invigilators: None.

Information for Candidates:

Numbers in brackets in the right margin (e.g. [5]) indicate maximum marks for each section of each question.

The following constants may be used:

electron charge : $e = 1.6 \times 10^{-19}$ C

permittivity of free space : $\varepsilon_o = 8.85 \times 10^{-12}$ F/m

relative permittivity of silicon : $\varepsilon_r = 12$

Planck's constant : $h = 6.63 \times 10^{-34}$ J s

Boltzmann's constant : $k = 1.38 \times 10^{-23}$ J/K

speed of light : $c = 3.0 \times 10^{8}$ m/s

The eigenvalue equations for transverse electric (TE) modes in a symmetric slab waveguide of thickness d are:

$\kappa = k_{1x} tan(k_{1x} d/2)$ and $\kappa = -k_{1x}\, cot(k_{1x} d/2)$

Dispersion Parameter approximation: At 3 dBm dispersion power penalty, the dispersion parameter can be approximated as: $BLD\sigma_\lambda = 0.2$.

1. You should attempt all parts of this question. Short answers only are required; there is no need to re-state the questions in your answer book, but you should show any calculations you use to arrive at your answers, state any assumptions or approximations made, and give a brief (one or two lines) explanation where appropriate. All parts have equal value.

[20]

a) An optical signal arrives at the end of a 200 km length of fibre with 0.1% of the intensity launched at the transmitter end. Calculate the fibre loss coefficient in dB/km.

b) A certain slab waveguide supports two TE modes, $m=0$ and $m=1$. For the $m=1$ mode, is the optical intensity higher at the centre of the guide, or at the core-cladding boundary? Give a brief explanation.

c) A certain slab waveguide supports 20 TE modes at a nominal wavelength $\lambda_o = 720$ nm. For an increased wavelength of $\lambda_o = 1300$ nm, would you expect the guide to support fewer, more or the same number of TE modes? Give a brief explanation.

d) If at some instant in time a 500 m length of optical fibre contains 7500 bits of propagating information, what is the bit rate of this signal?

e) Why is silicon not a suitable material for constructing detectors for optical systems operating at nominal wavelength 1650 nm?

f) Give a practical disadvantage of polarisation maintaining fibre in comparison to conventional single mode fibre.

g) Silica fibres generally have higher attenuation at 1.3 μm than at 1.5 μm wavelength. Which attenuation mechanism is primarily responsible for this higher attenuation and why?

h) A laser diode with quantum efficiency 0.80 has a slope efficiency of 1.16 W/A. Calculate the output (free space) wavelength.

i) Calculate the optimal thickness and refractive index of an anti-reflection coating acting between a glass of index 1.48 and air, for a free space wavelength of 0.75 μm.

j) Briefly explain why an avalanche photodiode usually has the avalanche region confined to the base of the depleted region.

2. A symmetric slab waveguide has core and cladding indices of n_1 = 1.46 and n_2 = 1.45 respectively. The guide is shown in Fig. 2.1. The core thickness d is 12 μm. A certain mode, at free-space wavelength 1.50 μm has an effective index of n' = 1.45264.

a) Define the boundary conditions required at the core-cladding interfaces for TE modes, with respect to the electric field distribution $E(x)$. Show that the effective index given above is consistent with these conditions.

[10]

b) Find the mode number m and sketch the corresponding field profile $E(x)$ for the mode having this value of n'.

[5]

c) What is the total number of TE modes supported by the guide at this wavelength?

[5]

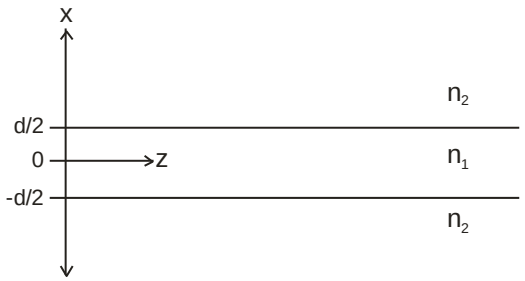


Figure 2.1 Slab waveguide

3. a) Write the equations that define the phase and group velocity of a propagating electromagnetic wave. What is the main physical significance of phase velocity?

[6]

b) At a given spectrum band, the refractive index of a sample of silica glass can be approximated by

$$n(\lambda_o) \cong D_o + D_1 \lambda_o^{-2} - D_2 \lambda_o^{2} \qquad (5.1)$$

where D_o = 1.41, D_1 = 0.002 μm^2 and D_2 = 0.003 μm^{-2}.

i) Calculate the phase and group velocities v_p and v_g in this material, as functions of D_o, D_1, D_2, and λ_o and the speed of light c. Which of the two is smaller at this spectrum band?

[6]

ii) Derive an expression for the phase change $\Delta\varphi$ experienced by a wave of nominal wavelength λ_o propagating a distance L.

[4]

iii) Derive an expression of wavelength at which dispersion is zero, as a function of D_1 and D_2. Calculate the wavelength for the values given above.

[4]

4. a) Describe and discuss the important attenuation mechanisms in optical fibres, and the influence these have in choice of operating wavelengths. Use diagrams and equations where appropriate.

[6]

b) An optical receiver can be implemented as a photodiode in series with a resistor, followed by a voltage amplifier. State the principal noise sources in this case, and draw a noise equivalent circuit for the receiver.

[4]

c) A certain optical link has a receiver with noise equivalent power (NEP) of 9.0 pW/√Hz, at the end of 100 km of fibre having attenuation of 0.4 dB/km at the operating wavelength of 1210 nm. The transmitted power is 5 mW and the photodetector quantum efficiency can be approximated as 1.

i) Determine whether shot noise or receiver noise is greater in this case;

[4]

ii) Considering only the dominant noise source as determined in (i), find the maximum bit rate for an optical *SNR* of 12;

[4]

iii) For the bit rate determined in (ii), find an approximate value for the maximum dispersion coefficient for this fibre above which the dispersion penalty would exceed 3 dB. Assume a transmitter spectral width σ_λ of 1.2 nm.

[2]

5. a) A silicon p-n photodiode (Fig. 5.1) has a depletion layer thickness of w, and p and n doping levels respectively of N_A and N_D, with $N_A = 5N_D = 4\times10^{20}$ m^{-3}. The quantities w_p and w_n are the depleted widths in the p and n regions respectively. A reverse bias voltage V_b is applied. Find an expression for the full depletion width w as a function of V_b, and the value of V_b for which w =5 μm.

[8]

b) Neglecting Fresnel reflection, find an expression for the quantum efficiency η of the photodiode of (a) if the total p region thickness h = 8 μm, and the absorption coefficient $\alpha = 0.866\times10^5$ m^{-1}. Hence find the value of V_b for which $\eta = 0.80$.

[6]

c) Give a typical approximate value for the saturation drift velocity of electrons in a semiconductor. For the photodiode of part (a) and an electron mobility in the silicon of 0.12 m^2/Vs, find the bias voltage for which the electron drift velocity is above this saturation value in half of the depleted region.

[6]

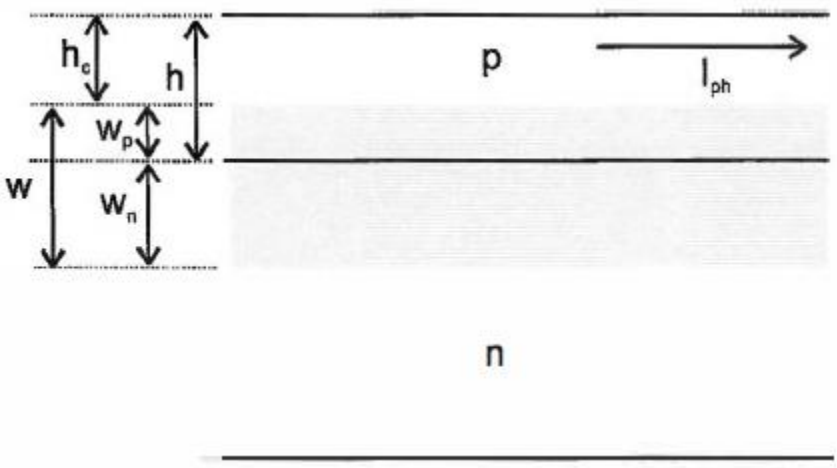


Figure 5.1: p-n photodiode.